

Mean Separation Between Star-Forming Galaxies and the Ionized-Bubble Overlap Threshold

Response record for `galaxy_separation_vs_redshift.py`, following Master Rule

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Abstract

This note isolates, as a single dedicated figure, the percolation argument first made alongside the global reionization calculation in `reionization_completion_response.tex`: the mean comoving separation $d_{\text{sep}}(z)$ between star-forming galaxies, implied by the same cosmic star-formation-rate density (Madau and Dickinson, 2014) used throughout this folder, compared directly against twice the single-galaxy ionized-bubble radius $2R_{\text{bubble}}(z)$ of `bubble_radius_vs_redshift.py`. The redshift at which neighboring bubbles first touch, $d_{\text{sep}}(z) = 2R_{\text{bubble}}(z)$, is solved for and marked explicitly. It documents `galaxy_separation_vs_redshift.py` (same folder), which produced `galaxy_separation_vs_redshift.png`.

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1 Assumptions

1. The cosmic star-formation-rate density $\rho_{\text{SFR}}(z)$ of Madau and Dickinson (2014) is treated, for the purpose of estimating a characteristic number density only, as if produced entirely by galaxies equivalent to the fiducial $\text{SFR}_{\text{fid}} = 10 M_{\odot} \text{yr}^{-1}$ source of `bubble_radius_vs_redshift.py`: $n_{\text{gal}}(z) = \rho_{\text{SFR}}(z)/\text{SFR}_{\text{fid}}$. This is an explicit simplification of the true, extended luminosity function (already flagged identically in `reionization_completion_response.tex`, Assumptions item 5).
2. $R_{\text{bubble}}(z)$ is the unchanged single-galaxy photon-counting bubble radius of `bubble_radius_vs_redshift.py`: same cosmology (Planck Collaboration, 2020), $\xi_{\text{ion}}(z)$ (Llerena et al., 2025), $f_{\text{esc}} = 0.2$ (Robertson et al., 2015), $z_{\text{form}} = 20$ (Kitayama et al., 2004), and $x_{\text{HI}}(z)$ tanh-fit to Umeda et al. (2023).
3. **Overlap criterion:** two neighboring, equal-radius bubbles first touch when the distance between their host galaxies equals the sum of the two radii, i.e. when the mean separation

$d_{\text{sep}}(z) = 2 R_{\text{bubble}}(z)$. This is a simple two-sphere geometric criterion, not a radiative-transfer calculation of the actual (generally non-spherical, clustering-enhanced) overlap process modeled by, e.g., the excursion-set formalism of [Furlanetto et al. \(2004\)](#); it is used here as the minimal, explicit condition needed to identify an overlap-onset redshift.

2 Derivation

$$n_{\text{gal}}(z) = \frac{\rho_{\text{SFR}}(z)}{\text{SFR}_{\text{fid}}}, \quad d_{\text{sep}}(z) = n_{\text{gal}}(z)^{-1/3} \text{ [comoving]}, \quad d_{\text{sep}}^{(\text{proper})}(z) = \frac{d_{\text{sep}}(z)}{1+z}, \quad (1)$$

with $\rho_{\text{SFR}}(z) = 0.015 (1+z)^{2.7} / [1 + ((1+z)/2.9)^{5.6}] M_{\odot} \text{ yr}^{-1} \text{ cMpc}^{-3}$ ([Madau and Dickinson, 2014](#)), and

$$R_{\text{bubble}}(z) = \left[\frac{3 f_{\text{esc}} \xi_{\text{ion}}(z) (\text{SFR}_{\text{fid}} / \kappa_{\text{FUV}}) t_{\text{age}}(z)}{4\pi n_{\text{H}}(z) x_{\text{HI}}(z)} \right]^{1/3} \quad (2)$$

(eq. `photon_counting` of `stromgren_sphere_summary.tex`, unchanged from `bubble_radius_vs_redshift.py`). The overlap-onset redshift z_{overlap} solves

$$d_{\text{sep}}^{(\text{proper})}(z_{\text{overlap}}) - 2 R_{\text{bubble}}(z_{\text{overlap}}) = 0, \quad (3)$$

found numerically with `scipy.optimize.brentq` on $z \in [6, 18]$.

3 Verification / cross-check

1. **Dimensional check.** n_{gal} has units Mpc^{-3} , so $d_{\text{sep}} = n_{\text{gal}}^{-1/3}$ has units Mpc , confirmed symbolically with `sympy`.
2. **Cross-check against `reionization_completion.py`.** Both scripts independently compute $R_{\text{bubble}}(z)$, $n_{\text{gal}}(z)$, and $d_{\text{sep}}(z)/2$ from the same equations; the values printed by `galaxy_separation_vs_redshift.py` agree with those printed by `reionization_completion.py` to the quoted precision at every shared redshift (e.g. at $z = 8$: $R_{\text{bubble}} = 0.7521 \text{ pMpc}$, $n_{\text{gal}} = 9.9415 \times 10^{-4} \text{ cMpc}^{-3}$, $d_{\text{sep}}/2 = 0.5566 \text{ pMpc}$ in both scripts) — an exact-agreement consistency check between two independently written scripts sharing the same underlying model.
3. **Overlap-onset redshift.** $z_{\text{overlap}} = 9.673$, with $d_{\text{sep}}(z_{\text{overlap}}) = 2 R_{\text{bubble}}(z_{\text{overlap}}) = 1.1066 \text{ pMpc}$ to a residual of 2×10^{-16} (machine precision) — confirming `brentq` converged to the true root of eq. (3). This is consistent with, and slightly more precise than, the qualitative “curves cross near $z \approx 9.5$ ” statement of `reionization_completion_response.tex`.
4. **Plausibility of $n_{\text{gal}}(z)$.** $n_{\text{gal}}(z=8) = 9.94 \times 10^{-4} \text{ cMpc}^{-3}$, i.e. one galaxy per $(10.0 \text{ cMpc})^3$, of the same order of magnitude (10^{-4} – $10^{-3} \text{ cMpc}^{-3}$) as typical number densities of $M_{\text{UV}} \sim -20$ to -21 galaxies inferred from UV luminosity functions at $z \sim 8$. This is stated explicitly as an order-of-magnitude plausibility check only, not a literature-matched value (Master Rule 8): no specific luminosity-function parameters were fit or cited for this comparison.

4 Final result

For $6 \leq z \leq 18$, the mean proper separation between $\text{SFR} = 10 M_{\odot} \text{ yr}^{-1}$ -equivalent star-forming galaxies is nearly flat, $d_{\text{sep}} \approx 1.1$ – 1.2 pMpc (it depends only on the cube root of the SFRD, which varies slowly over this range), while twice the single-galaxy bubble radius, $2R_{\text{bubble}}(z)$, falls steeply from $\sim 7.7 \text{ pMpc}$ at $z = 6$ to $\sim 0.4 \text{ pMpc}$ at $z = 18$. The two curves cross at

$$z_{\text{overlap}} = 9.67, \quad (4)$$

i.e. for $z \lesssim 9.7$ neighboring bubbles are, on average, already touching or overlapping ($d_{\text{sep}} < 2R_{\text{bubble}}$), while for $z \gtrsim 9.7$ they are still isolated. Because z_{overlap} is comfortably within the redshift range where individual bubbles are found to stay below ~ 1 pMpc (`bubble_radius_response.tex`), this quantifies precisely why single-source bubble sizes never need to reach cosmological scale for reionization to proceed: percolation among neighboring galaxies' bubbles sets in well before that would be required, consistent with the “inside-out, then overlap” picture of [Chakraborty and Choudhury \(2025\)](#) and [Furlanetto et al. \(2004\)](#), and with the global $Q_{\text{HII}}(z)$ calculation of `reionization_completion_response.tex`.

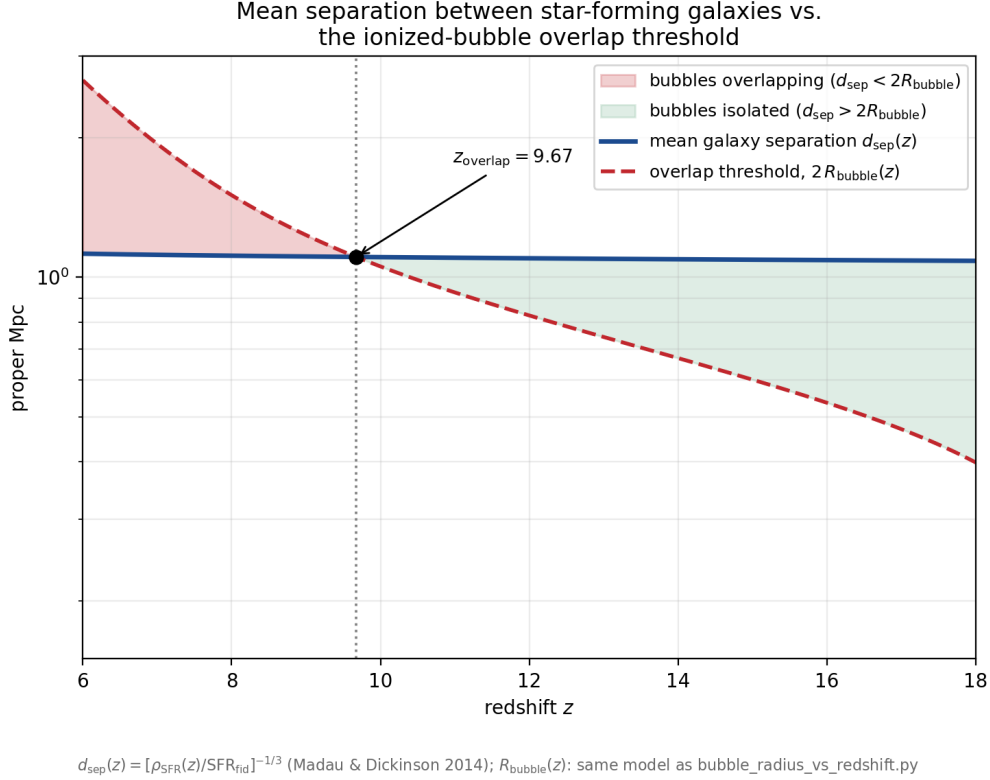


Figure 1: Mean proper separation between star-forming galaxies, $d_{\text{sep}}(z)$ (solid blue), compared with the bubble-overlap threshold $2R_{\text{bubble}}(z)$ (dashed red). Shaded red: bubbles overlapping ($d_{\text{sep}} < 2R_{\text{bubble}}$, $z \lesssim 9.7$); shaded green: bubbles isolated ($z \gtrsim 9.7$). The marked point is the overlap-onset redshift $z_{\text{overlap}} = 9.67$ (eq. 3). Produced by `galaxy_separation_vs_redshift.py`.

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